

AN UPDATED MODEL FOR $O_2(a^1\Delta_g)$ CONCENTRATIONS IN THE MESOSPHERE AND LOWER THERMOSPHERE AND IMPLICATIONS FOR REMOTE SENSING OF OZONE AT 1.27 μm

Martin G. Mlynczak

Atmospheric Sciences Division, NASA Langley Research Center, Hampton, Virginia

Susan Solomon

NOAA Aeronomy Laboratory, Boulder, Colorado

Daphne S. Zaras

Science Applications International Corporation, Hampton, Virginia

Abstract. An updated kinetic model for the calculation of daytime steady state $O_2(a^1\Delta_g)$ concentrations in the mesosphere and lower thermosphere is presented. In addition to the usual source from ozone photolysis in the Hartley band, energy transfer from $O(^1D)$ produced by photolysis of molecular oxygen in the Lyman alpha and Schumann-Runge continuum spectral intervals is shown to be a significant source of $O_2(a^1\Delta_g)$ as well. Numerous changes and additions to the basic $O_2(a^1\Delta_g)$ kinetic model have been included. These new sources and kinetic parameters imply that lower mesospheric ozone concentrations as inferred by near-infrared emission techniques are to be decreased by as much as 10% while lower thermospheric ozone concentrations are to be increased by as much as 20%.

1. Introduction

Molecular oxygen is formed in its lowest-lying electronic state, $O_2(a^1\Delta_g)$, directly upon the photolysis of ozone in the Hartley band. $O_2(a^1\Delta_g)$ is also produced indirectly by energy transfer from $O(^1D)$, which is another Hartley band photolysis product. In the mesosphere and lower thermosphere, $O_2(a^1\Delta_g)$ is not completely quenched by collisions and, in fact, radiates quite strongly in the infrared atmospheric band $O_2(a^1\Delta_g - X^3\Sigma_g)$ at 1.27 μm . A determination of the local $O_2(a^1\Delta_g)$ volume emission rate as a function of altitude can therefore be used as a proxy from which ozone concentrations can be inferred. This technique was employed in the Solar-Mesosphere Explorer (SME) satellite experiment in order to derive ozone concentrations in the terrestrial mesosphere [Thomas et al., 1984]. The key to interpreting measurements of $O_2(a^1\Delta_g)$ in terms of ozone concentration depends not only on an accurate knowledge of the kinetic and spectroscopic parameters describing the production and quenching of $O_2(a^1\Delta_g)$ produced by ozone photolysis but also on a detailed accounting of all sources of $O_2(a^1\Delta_g)$ in the ozone retrieval/inference process. Failure to account for additional

$O_2(a^1\Delta_g)$ arising from processes other than ozone photolysis will result in a systematic overestimation of the inferred ozone abundance. Errors in the kinetic and spectroscopic parameters used to calculate the $O_2(a^1\Delta_g)$ abundance also lead to systematic errors in the inferred ozone.

The photolysis of molecular oxygen in the Schumann-Runge continuum (SRC) and in the Lyman alpha ($Ly \alpha$) spectral regions also results in the production of $O(^1D)$ atoms. As with $O(^1D)$ produced by photolysis of ozone, some of the internal energy of O_2 photolysis-produced $O(^1D)$ is transferred to molecular oxygen during collisions, ultimately producing $O_2(a^1\Delta_g)$. In this paper we evaluate the contribution to the total $O_2(a^1\Delta_g)$ concentration resulting from the photolysis of molecular oxygen. We will show that $Ly \alpha$ photolysis of O_2 may contribute up to 15% of the total $O_2(a^1\Delta_g)$ concentration in the vicinity of 80 km, while $Ly \alpha$ and SRC photolysis may contribute up to 20% of the $O_2(a^1\Delta_g)$ at 92 km. Above about 95 km, $O_2(a^1\Delta_g)$ produced subsequent to O_2 photolysis in the SRC becomes the dominant source of $O_2(a^1\Delta_g)$ in that region.

While past experiments such as SME have been successful in defining the general behavior of mesospheric ozone, future experiments are being conceived to address issues such as global change in the mesosphere. In fact, the thermosphere-ionosphere-mesosphere energetics and dynamics (TIMED) mission currently under consideration by the National Aeronautics and Space Administration (NASA) has a goal of creating a baseline data set from which future changes could be judged [NASA, 1991]. Given that the mesosphere is subject to large natural variations and that anthropogenic variations may occur on the timescale of decades, it is imperative that the new data sets be both precise and accurate, to perhaps the 10% level. Furthermore, for accurate determination of the mesospheric heat budget, knowledge of the ozone concentration to $\sim 10\%$ is also desired, since the absorption of solar ultraviolet radiation by ozone is the primary radiative forcing mechanism in the mesosphere. To achieve such accuracies, rate constants used in the statistical equilibrium model must be known quite well and sources of $O_2(a^1\Delta_g)$ emission arising from processes other than ozone photolysis must be very well characterized.

In section 2 we outline the kinetic model used in the calculation of $O_2(a^1\Delta_g)$ concentrations. Numerous additions and updates to the $O_2(a^1\Delta_g)$ kinetic model have been included

Copyright 1993 by the American Geophysical Union.

Paper number 93JD01478.

0148-0227/93/93JD-01478 \$05.00

over that used in the analysis of SME data. All calculations presented here are for solar maximum conditions. In addition, we solve the system of equations under the assumption of midday steady state, but we note that the long lifetime (65 min) of $O_2(a^1\Delta_g)$ implies that transport of $O_2(a^1\Delta_g)$ might be important under some instances [e.g., Sica, 1991]. In section 3 we compare the magnitude of $O_2(a^1\Delta_g)$ produced subsequent to O_2 photolysis to that produced directly by O_3 photolysis. This comparison is done using both measured and modeled ozone concentrations and will demonstrate the importance of these additional sources. We also evaluate the effect that uncertainties in the quantum yield of $O(^1D)$ production at Ly α and in the transfer efficiency of $O(^1D)$ energy to O_2 have on the $O_2(a^1\Delta_g)$ produced subsequent to O_2 photolysis. In section 4 we examine the impact of the new kinetic model on existing ozone databases developed from measurements of $O_2(a^1\Delta_g)$ emission. In section 4 we illustrate the changes that can occur in a data set derived from measurements of nonlocal thermodynamic equilibrium emission if rate constants used in the statistical equilibrium model are adjusted as a consequence of new determinations or evaluations. Finally, results are summarized in section 5, including a discussion of the likely dependence on solar activity.

2. The Kinetic Model

The model we use for production and quenching of $O_2(a^1\Delta_g)$ is given in Table 1. Except as noted in Table 1, we use rate constants from the latest panel recommendations and evaluations. Ozone is dissociated in the Hartley band, yielding $O_2(a^1\Delta_g)$ and $O(^1D)$. The quantum yield for this process is 0.9. We also include the production of $O(^1D)$ resulting from the photolysis of O_2 in the SRC and Ly α spectral intervals. $O(^1D)$ is important as it transfers some of its energy to O_2 ,

forming $O_2(b^1\Sigma_g)$, which may be quenched to $O_2(a^1\Delta_g)$. The quantum yield for the production of $O(^1D)$ in the SRC is 1.0, while the yield is estimated to be 0.44 at 1216.5 Angstroms [Lee et al., 1977]. However, the quantum yield in the Ly α spectral region (ϕ_b in Table 1) is a very strong function of wavelength. The value reported by Lee et al. [1977] is at a discrete wavelength and may not represent a mean value over the Ly α absorption feature. Consequently, the correct value for use in atmospheric studies may be larger than the reported value of 0.44, which is the minimum value reported in that spectral region. The $O_2(b^1\Sigma_g)$ state is also excited directly upon absorption of solar radiation at 762 nm and may be quenched to the $O_2(a^1\Delta_g)$ state. The solar excitation rate is calculated using band strengths from the HITRAN [Rothman et al., 1987] spectroscopic database and solar flux values at 762 nm from World Meteorological Organization [1981]. We have found the rate for this process to be virtually insensitive to variations in kinetic temperature. $O(^1D)$ is quenched during collisions with N_2 and O_2 . As mentioned above, collisions with O_2 result in the formation of the $O_2(b^1\Sigma_g)$ state. A critical parameter is the efficiency of energy transfer from $O(^1D)$ to O_2 (ϕ_c in Table 1) in the formation of $O_2(b^1\Sigma_g)$. Lee and Slanger [1978] report a value of 0.77 ± 0.23 , implying a transfer efficiency that lies between 0.54 and 1.0. $O_2(b^1\Sigma_g)$ is quenched upon collisions with N_2 , O_2 , CO_2 , O_3 , and O , with CO_2 responsible for about 8% of the total quenching. $O_2(b^1\Sigma_g)$ radiates in the atmospheric band with an inverse radiative lifetime of 0.0758 s^{-1} . $O_2(a^1\Delta_g)$ is quenched upon collisions with O_2 , N_2 , and O and radiates in the infrared atmospheric band at $1.27 \mu\text{m}$ with an inverse radiative lifetime of $2.58 \times 10^{-4} \text{ s}^{-1}$. Quenching of $O_2(a^1\Delta_g)$ by atomic oxygen is important in the lower thermosphere as at 10^{-3} mbar ($\sim 92 \text{ km}$) it is responsible for about 20% of the removal of $O_2(a^1\Delta_g)$, while quenching by O_2 accounts for only about 5%. The processes listed in Table 1 are illustrated in Figure 1.

TABLE 1. Kinetic and Spectroscopic Processes Used in the Evaluation of the $O_2(a^1\Delta_g)$ Concentration

Process	Rate	Efficiency or Quantum Yield	Reference
$O_3 + h\nu \rightarrow O_2(^1\Delta) + O(^1D)$	Hartley band, $J_\infty = 8.1 \times 10^{-3}$	$\phi_a = 0.9$	JPL [1992]
$O_2 + h\nu \rightarrow O(^3P) + O(^1D)$	Ly- α , $J_\infty = 4.5 \times 10^{-9}$	$0.44 < \phi_b < 1.0$	see text
$O_2 + h\nu \rightarrow O(^3P) + O(^1D)$	SRC, $J_\infty = 6.3 \times 10^{-6}$	1.0	see text
$O_2 + h\nu (762 \text{ nm}) \rightarrow O_2(^1\Sigma)$	5.35×10^{-9}		HITRAN, 1987; see text
$O(^1D) + N_2 \rightarrow O(^3P) + N_2$	$1.8 \times 10^{-11} e^{110/T}$		JPL [1992]
$O(^1D) + O_2 \rightarrow O_2(^1\Sigma) + O(^3P)$	$3.2 \times 10^{-11} e^{70/T}$	$0.54 < \phi_c < 1.0$	JPL [1992]; see text
$O(^1D) + O_2 \rightarrow O_2 + O(^3P)$	$3.2 \times 10^{-11} e^{70/T}$	$1 - \phi_c$	JPL [1992]; see text
$O_2(^1\Delta) + O_2 \rightarrow O_2 + O_2$	$3.6 \times 10^{-18} e^{-220/T}$		JPL [1992]
$O_2(^1\Delta) + N_2 \rightarrow O_2 + N_2$	1.0×10^{-20}		JPL [1992]
$O_2(^1\Delta) + O \rightarrow O_2 + O$	1.3×10^{-16}		Clark and Wayne [1969]
$O_2(^1\Delta) \rightarrow O_2 + h\nu (1.27 \mu\text{m})$	2.58×10^{-4}		HITRAN, 1987
$O_2(^1\Sigma) + N_2 \rightarrow O_2(^1\Delta) + N_2$	2.1×10^{-15}		JPL [1992]
$O_2(^1\Sigma) + CO_2 \rightarrow O_2(^1\Delta) + CO_2$	4.2×10^{-13}		JPL [1992]
$O_2(^1\Sigma) + O_3 \rightarrow O_2(^1\Delta) + O_3$	2.2×10^{-11}		JPL [1992]
$O_2(^1\Sigma) + O \rightarrow O_2(^1\Delta) + O$	8.0×10^{-14}		JPL [1992]
$O_2(^1\Sigma) + O_2 \rightarrow O_2(^1\Delta) + O_2$	3.9×10^{-17}		JPL [1992]
$O_2(^1\Sigma) \rightarrow O_2 + h\nu (762 \text{ nm})$	0.0758		HITRAN, 1987; see text

Oxygen Dayglow Production Mechanism

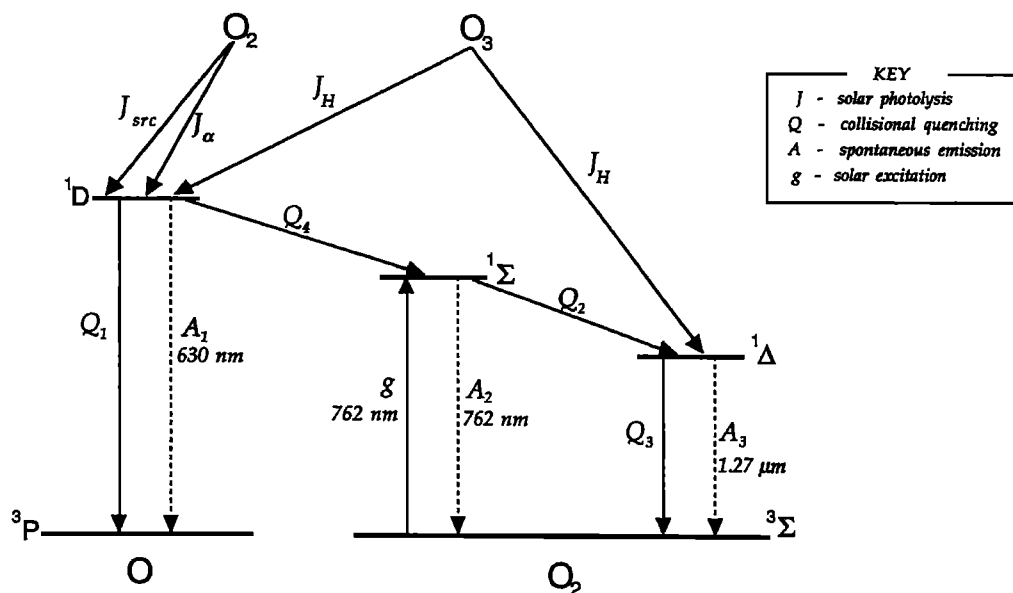


Fig. 1. Illustration of the processes listed in Table 1 which result in the generation of airglow at 1.27 μm and 762 nm.

The model presented in Table 1 is similar to that used by Thomas et al. [1984] (also see WMO [1988]), with several important differences. First, the Hartley band photolysis rate is 10% smaller than used in SME and is calculated using solar fluxes and cross sections from Jet Propulsion Laboratory [1992]. Second, we consider two additional indirect sources of $O_2(a^1\Delta_g)$ provided by energy transfer from $O(^1D)$. As discussed later, these two sources can be quite significant in the upper mesosphere and lower thermosphere. However, uncertainties in the quantum yield at Ly α wavelengths and in the transfer efficiency of $O(^1D)$ energy to O_2 limit the ability to decisively calculate the contributions of energy transfer from $O(^1D)$ to the overall $O_2(a^1\Delta_g)$ concentration. We will look at the extrema of $O_2(a^1\Delta_g)$ produced by energy transfer from $O(^1D)$ resulting from the uncertainties in these yields in section 3. We also include quenching of $O_2(a^1\Delta_g)$ by O and N_2 and quenching of $O_2(b^1\Sigma_g)$ by O, O_2 , CO_2 , and O_3 . The correct temperature dependence of the quenching of $O(^1D)$ by N_2 and O_2 is incorporated in our model. Finally, the rate of radiative excitation of $O_2(b^1\Sigma_g)$ at 762 nm is 15% smaller than used by SME. For comparison the SME kinetic model is given in Table 2.

There is one key point of uncertainty in the $O_2(a^1\Delta_g)$ model. The product state resulting from the quenching of $O_2(b^1\Sigma_g)$, which is assumed to be $O_2(a^1\Delta_g)$, has apparently never been measured directly in the gas phase in the laboratory, at least for collisions with N_2 , the primary atmospheric quencher of $O_2(b^1\Sigma_g)$. Thomas et al. [1984] assumed that the product state is $O_2(a^1\Delta_g)$ based on work of Ogryzlo and Thrush [1974] who studied energy transfer between H_2O and $O_2(b^1\Sigma_g)$ and between CO_2 and $O_2(b^1\Sigma_g)$ and found no evidence to indicate that $O_2(a^1\Delta_g)$ was not the product state. Braithwaite et al. [1976] state that the products of quenching of $O_2(b^1\Sigma_g)$ by N_2 are $O_2(a^1\Delta_g, v = 2)$ and

$N_2(v = 1)$ based on trajectory calculations. The laboratory measurements of Knickelbein et al. [1987] showed that the fraction of collisions between $O_2(b^1\Sigma_g)$ and O_2 which result in the population of the $O_2(a^1\Delta_g)$ state is 1.1 ± 0.2 . Similarly, the measurements reported by Wildt et al. [1988] show that $O_2(a^1\Delta_g)$ is produced at least 90% of the time during collisions between $O_2(b^1\Sigma_g)$ and several gases, although N_2 was not included in their measurements. The only other electronic state below $O_2(b^1\Sigma_g)$ is the ground $O_2(X^3\Sigma_g)$ state. If $O_2(b^1\Sigma_g)$ is quenched (wholly or partially) to the ground state rather than to $O_2(a^1\Delta_g)$, then the contribution to the $O_2(a^1\Delta_g)$ abundance from energy transfer from $O(^1D)$ produced by O_3 photolysis and O_2 photolysis, as well as $O_2(a^1\Delta_g)$ resulting from solar excitation of $O_2(b^1\Sigma_g)$, will be reduced accordingly. For the purposes of this study we will assume that all collisions involving $O_2(b^1\Sigma_g)$ result in the formation of $O_2(a^1\Delta_g)$, but we note that these processes need to be confirmed by direct laboratory measurement in the gas phase over the range of kinetic temperatures encountered in the mesosphere.

3. Magnitudes of Additional Sources of $O_2(a^1\Delta_g)$ From O_2 Photolysis

We have solved the system of kinetic equations listed in Table 1 under steady state conditions, separately calculating the contributions to the $O_2(a^1\Delta_g)$ concentration from O_3 photolysis, O_2 photolysis, and solar excitation of O_2 at 762 nm. In these calculations the quantum yield of Ly α photolysis and the $O(^1D)$ - $O_2(b^1\Sigma_g)$ transfer efficiencies are assumed equal to 1.0, thus providing the maximum possible $O_2(a^1\Delta_g)$ from O_2 photolysis. We discuss the effect of nonunit yields and efficiencies later in this section. Ozone concentrations, kinetic temperatures, and O_2 photolysis rates as functions of pressure are initially taken from the two-

TABLE 2. Kinetic and Spectroscopic Processes Used in the Evaluation of SME $O_2(a^1\Delta_g)$ Measurements

Process	Rate	Efficiency or Quantum Yield	Reference
$O_3 + h\nu \rightarrow O_2(^1\Delta) + O(^1D)$	Hartley band, $J_{\infty} = 9.0 \times 10^{-3}$	$\phi_a = 0.9$	WMO [1988]
$O_2 + h\nu (762 \text{ nm}) \rightarrow O_2(^1\Sigma)$	6.3×10^{-9}		Thomas et al. [1984]
$O(^1D) + N_2 \rightarrow O(^3P) + N_2$	$2.0 \times 10^{-11} e^{-107/T}$		Thomas et al. [1984]
$O(^1D) + O_2 \rightarrow O_2(^1\Sigma) + O(^3P)$	$2.9 \times 10^{-11} e^{-67/T}$		Thomas et al. [1984]
$O_2(^1\Delta) + O_2 \rightarrow O_2 + O_2$	$2.22 \times 10^{-18} (T/300)^{0.78}$		Thomas et al. [1984]
$O_2(^1\Delta) \rightarrow O_2 + h\nu (1.27 \mu\text{m})$	2.58×10^{-4}		Thomas et al. [1984]
$O_2(^1\Sigma) + N_2 \rightarrow O_2(^1\Delta) + N_2$	2.0×10^{-15}		Thomas et al. [1984]
$O_2(^1\Sigma) \rightarrow O_2 + h\nu (762 \text{ nm})$	0.085		Thomas et al. [1984]

dimensional model of Garcia and Solomon [1983, 1985].

Using our calculated values of $O_2(a^1\Delta_g)$ produced by each source, we evaluate the ratio $\mathcal{R}$ defined as

$$\mathcal{R} = \frac{O_2(a^1\Delta_g)_{\text{Ly}\alpha} + O_2(a^1\Delta_g)_{\text{SRC}}}{O_2(a^1\Delta_g)_{\text{O}_3}} \quad (1)$$

where $O_2(a^1\Delta_g)_{\text{Ly}\alpha}$ and $O_2(a^1\Delta_g)_{\text{SRC}}$ are the $O_2(a^1\Delta_g)$ concentrations produced by energy transfer from $O(^1D)$ that is created by O_2 photolysis at Ly α and SRC wavelengths. $O_2(a^1\Delta_g)_{\text{O}_3}$ is the $O_2(a^1\Delta_g)$ concentration resulting from the photolysis of ozone and includes the contributions from the direct production of $O_2(a^1\Delta_g)$ in the Hartley band and from the indirect production of $O_2(a^1\Delta_g)$ as a result of $O(^1D)$ formed by Hartley band photolysis.

Figure 2 shows a plot of $\mathcal{R}$ extending from the upper mesosphere into the lower thermosphere (65–107 km) at 25° S latitude at northern hemisphere winter solstice. The values are plotted as a function of $\log_{10}$ pressure (in millibars). The geometric altitudes (in kilometers) are approximate and correspond to the altitudes for pressure surfaces listed in the 1976 U.S. Standard Atmosphere. The solid curve in Figure 2 is the value of $\mathcal{R}$, while the dashed curves indicate the separate contributions to $\mathcal{R}$ from Ly α photolysis and from SRC photolysis. Below 85 km, Ly α is the dominant source of $O_2(a^1\Delta_g)$ produced by photolysis of O_2 , providing as much as 14% of the $O_2(a^1\Delta_g)$ as produced by the photolysis of ozone in the vicinity of 80 km. The SRC is not an important source of $O_2(a^1\Delta_g)$ below 85 km because it is effectively absorbed at higher altitudes. However, the SRC rapidly becomes an important source of $O_2(a^1\Delta_g)$ above 85 km and it becomes the dominant source of $O_2(a^1\Delta_g)$ above 97 km, exceeding that provided by O_3 photolysis (i.e., $\mathcal{R}$ is greater than 1 above ~97 km).

In principle, the O_2 photolysis-induced sources of $O_2(a^1\Delta_g)$ must be removed from measurements of the $O_2(a^1\Delta_g)$ volume emission rate prior to converting those measurements into ozone concentrations. The value of $\mathcal{R}$ represents the amount by which the ozone concentration would be overestimated if the additional sources of $O_2(a^1\Delta_g)$ were not accounted for in a retrieval, assuming that the remaining $O_2(a^1\Delta_g)$ sources and kinetics are known exactly. Therefore from Figure 2, failure to account for the O_2 photolysis-induced $O_2(a^1\Delta_g)$ would result in an overestimation of the ozone concentration by ~10% at 75 km, ~13% at 80 km, ~20% at 92 km, and ~100% at 97 km.

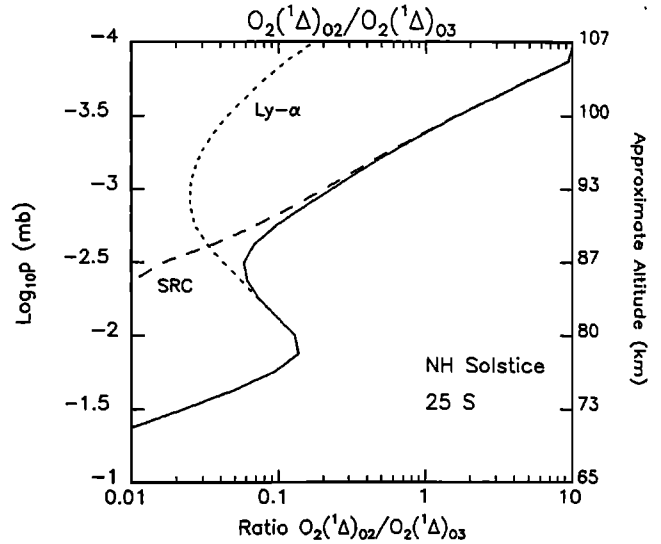


Fig. 2. The parameter $\mathcal{R}$ calculated with model ozone profiles at 25° S latitude during northern hemisphere winter. The dashed curves show the individual contributions to $\mathcal{R}$ from photolysis at Ly α and Schumann-Runge continuum wavelengths.

The results shown in Figure 2 are indicative of the contributions to the $O_2(a^1\Delta_g)$ concentrations at one latitude. It is important to determine the magnitude of the O_2 photolysis contributions to the $O_2(a^1\Delta_g)$ concentration as a function of latitude and season. It is expected that the value of $\mathcal{R}$ (and hence the effect on ozone retrievals) will vary with latitude and season because the ozone concentration and the O_2 photolysis rates vary in time and space. Because of the vertical gradient, we examine the value of $\mathcal{R}$ in two separate altitude regions, 65–87 km and 87–96 km.

Figures 3 and 4 show the values of $\mathcal{R}$ as a function of latitude and altitude during northern hemisphere winter solstice. These figures clearly show the significant contributions to the $O_2(a^1\Delta_g)$ concentrations from O_2 photolysis. The value of $\mathcal{R}$ is quite latitude dependent, implying that failure to account for the O_2 contribution to the $O_2(a^1\Delta_g)$ concentration would result not only in an overestimation of the ozone concentration but also in spurious temporal and latitudinal variations. The maximum value of $\mathcal{R}$ (15%) at 60° S at ~80 km in Figure 3 is a combination of the cold summer mesosphere which implies a larger O_2 density (and hence more $O_2(a^1\Delta_g)$ generation by Ly α photolysis on a

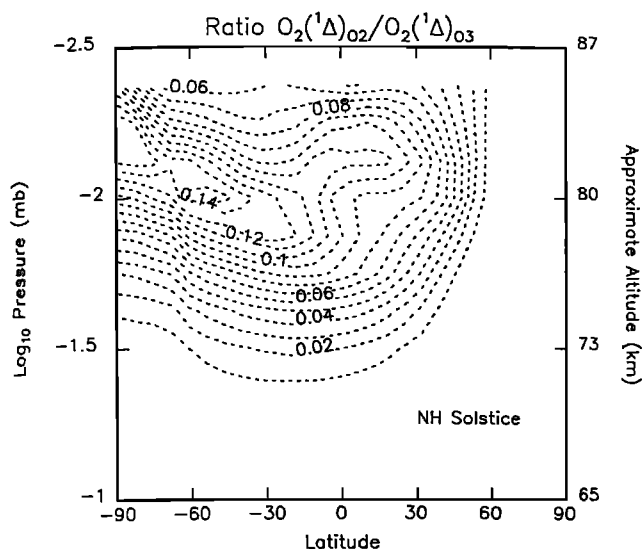


Fig. 3. The parameter R in the upper mesosphere calculated with model ozone profiles as a function of latitude during northern hemisphere winter.

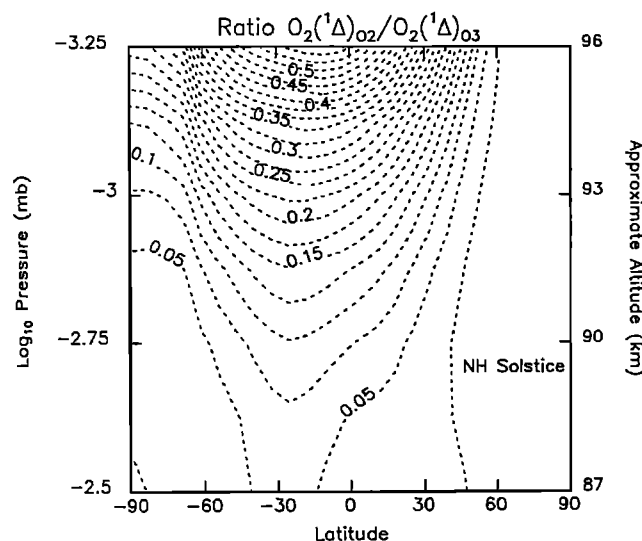


Fig. 4. The parameter R in the lower thermosphere calculated with model profiles as a function of latitude during northern hemisphere winter.

pressure surface) coupled with relatively lower model ozone on the same pressure surface. This effect illustrates the need for simultaneous measurements of kinetic temperature and pressure in addition to the measurement of $O_2(a^1\Delta_g)$ emission at $1.27 \mu\text{m}$ to correctly infer the ozone concentration. Figure 4 primarily shows the effect of SRC in rapidly increasing the amount of $O_2(a^1\Delta_g)$ produced by O_2 photolysis relative to that produced by O_3 photolysis in the lower thermosphere.

Figures 5 and 6 show the values of R at equinox. The results are qualitatively similar to those at solstice, with a potentially significant and latitudinally dependent value of R . In the upper mesosphere (Figure 5), O_2 photolysis contributes a maximum of 8 to 9% to R . The value of R is generally lower than at solstice due to the higher ozone concentrations. Figure 6 serves to again demonstrate the rapid increase in the

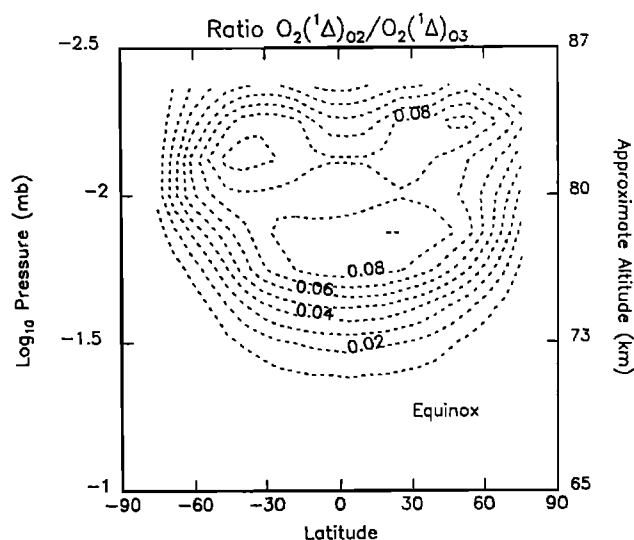


Fig. 5. The parameter R in the upper mesosphere calculated with model profiles as a function of latitude at equinox.

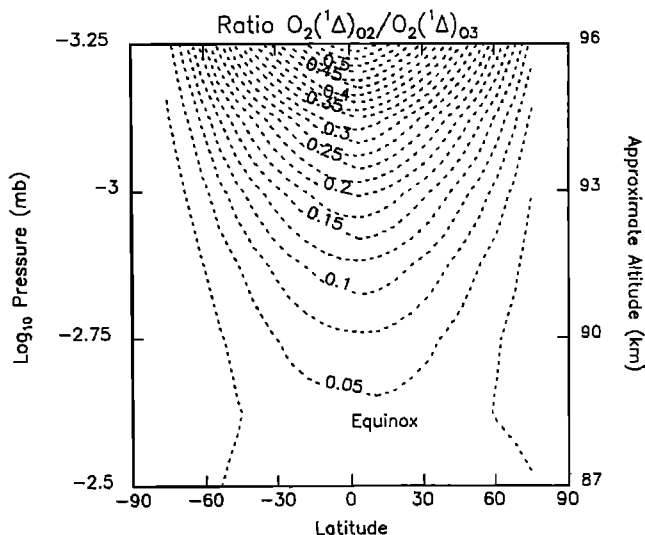


Fig. 6. The parameter R in the lower thermosphere calculated with model profiles as a function of latitude at equinox.

contribution to R from SRC photolysis above 85 km. The latitudinal dependence at equinox above 85 km is primarily due to variations in the SRC photolysis rate with solar zenith angle.

We recognize that modeled and measured ozone concentrations in the mesosphere are frequently not in agreement, with model ozone concentrations generally being less than reported by measurement. If the mesospheric ozone concentrations are larger than used in our studies reported above, we will have overestimated the ratio R and, consequently, the importance of O_2 photolysis sources of $O_2(a^1\Delta_g)$. To assess the value of R using atmospheric data, we have calculated R using March and December 1982 monthly mean SME data from the near-infrared spectrometer (NIRS). However, we use the SME forward model (Table 2) to calculate the $O_2(a^1\Delta_g)$ observed by the NIRS together with our forward model (i.e., Table 1) calculations of $O_2(a^1\Delta_g)$ concentrations resulting from O_2 photolysis. Because the

SME-observed $O_2(a^1\Delta_g)$ concentration also has contributions resulting from solar excitation of $O_2(b^1\Sigma_g)$, we define R_{SME} as

$$R_{SME} = \frac{O_2(^1\Delta)_{SRC} + O_2(^1\Delta)_{Ly\alpha}}{O_2(^1\Delta)_{SME} - O_2(^1\Delta)_{SRC} - O_2(^1\Delta)_{Ly\alpha} - O_2(^1\Delta)_{sun}} \quad (2)$$

where $O_2(^1\Delta)_{SME}$ is the total $O_2(a^1\Delta_g)$ concentration obtained using the SME forward model with an SME ozone profile. $O_2(^1\Delta)_{sun}$ is the $O_2(a^1\Delta_g)$ concentration due to solar excitation of $O_2(b^1\Sigma_g)$ followed by quenching to $O_2(a^1\Delta_g)$, calculated with the kinetics given in Table 1. The denominator in (2) represents the SME-observed $O_2(a^1\Delta_g)$ due to ozone photolysis alone, while the numerator represents the O_2 photolysis induced $O_2(a^1\Delta_g)$ concentration.

The values R_{SME} are shown in Figures 7 and 8 for solstice (December) and equinox (March) conditions, respectively. The upper altitude limit of the SME data is ~ 92 km (1.33×10^{-3} mbar). We have again assumed unit quantum yields of $O(^1D)$ from O_2 photolysis at Ly α and a unit transfer efficiency of $O(^1D)$ energy to O_2 . The results are quantitatively and qualitatively similar to those shown in Figures 3 through 6. For solstice conditions, there is a peak in the value of R_{SME} near 10^{-2} mbar at midlatitudes in the southern hemisphere of just under 7%, compared with the 15% peak value of R calculated with the model ozone profiles. At equinox the peak value of R_{SME} near 10^{-2} mbar is just under 6% compared with the R value of almost 9% obtained using the model ozone concentrations. The values of R_{SME} in the upper mesosphere are generally less than the values of R calculated with the Garcia and Solomon model ozone because of the larger ozone amounts found in the SME data. However, at the highest altitudes reported in the SME database (85–92 km), the values for R and R_{SME} are very nearly the same. Figure 9 compares the parameter R and R_{SME} at the equator at solstice. The largest discrepancy in the value of R exists between 75 and 85 km with reasonable agreement above and below those altitudes. Figure 9 demonstrates that the $O_2(a^1\Delta_g)$ produced by O_2 photolysis is a significant portion of the total $1.27\text{-}\mu\text{m}$ radiance if the ozone concentrations are equal to or less than those found in the SME data set, assuming that both the quantum yield for $O(^1D)$ at Ly α and the transfer efficiency for $O(^1D)$ to O_2 are equal to 1.0.

The values of R and R_{SME} are calculated assuming unit quantum yield for the generation of $O(^1D)$ by O_2 photolysis at Ly α and a unit transfer efficiency for the excitation of $O_2(b^1\Sigma_g)$ in collisions between O_2 and $O(^1D)$, thus giving the maximum possible values for R and R_{SME} . Lee et al. [1977] report a quantum yield of $O(^1D)$ from O_2 photolysis at Ly α of 0.44 while Lee and Slanger [1978] report that the $O(^1D)$ – O_2 transfer efficiency is 0.77 ± 0.23 , implying a minimum value of 0.54 and a maximum value of 1.0. As shown by Lee et al. [1977], the quantum yield of $O(^1D)$ is a very strong function of wavelength in the vicinity of 1216.5 Angstroms. Furthermore, the quantum yield was measured at a discrete wavelength over a relatively small portion of the O_2 Ly α absorption feature. What is required for our calculations is the quantum yield as a function of wavelength over the Ly α line, which can be convolved with the solar flux (F_v) and O_2 cross sections (σ_v) to determine a weighted mean quantum yield, i.e.,

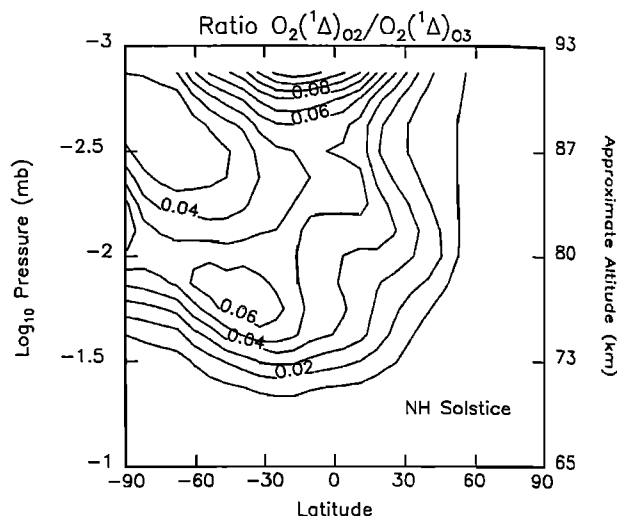


Fig. 7. The parameter R_{SME} in the upper mesosphere and lower thermosphere calculated with model profiles as a function of latitude during northern hemisphere winter.

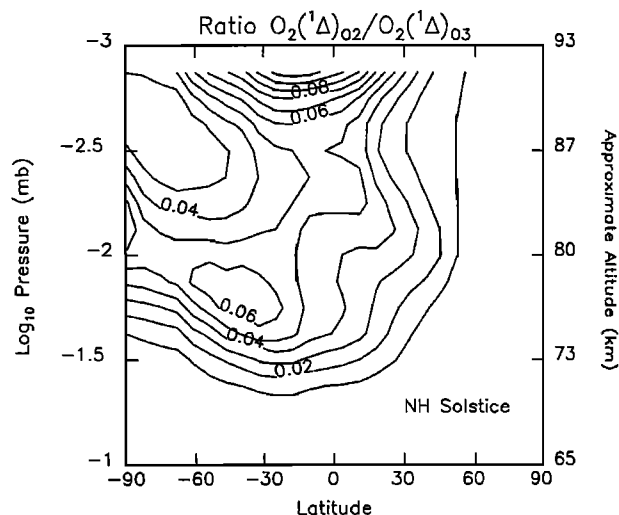


Fig. 8. The parameter R_{SME} in the upper mesosphere and lower thermosphere calculated with model profiles as a function of latitude at equinox.

$$\bar{\phi} = \frac{\int \phi_v F_v \sigma_v dv}{\int F_v \sigma_v dv} \quad (3)$$

The mean value of ϕ may be larger or smaller than the value of 0.44 reported by Lee et al. [1977].

To estimate the effects of nonunit $O(^1D)$ quantum yields from Ly α photolysis of O_2 and nonunit transfer efficiencies of $O(^1D)$ to O_2 , we have calculated both R and R_{SME} using a 0.44 $O(^1D)$ quantum yield and a 0.54 $O(^1D)$ – O_2 transfer efficiency, values which should result in the calculation of the minimum expected value of both R and R_{SME} . It should be noted that the 0.54 transfer efficiency also reduces the $O_2(a^1\Delta_g)$ resulting from O_3 photolysis. Figures 10 and 11 show values of R and R_{SME} (respectively) calculated assuming unit quantum yields and transfer efficiencies (solid curve) and values calculated with the 0.44 quantum yield and

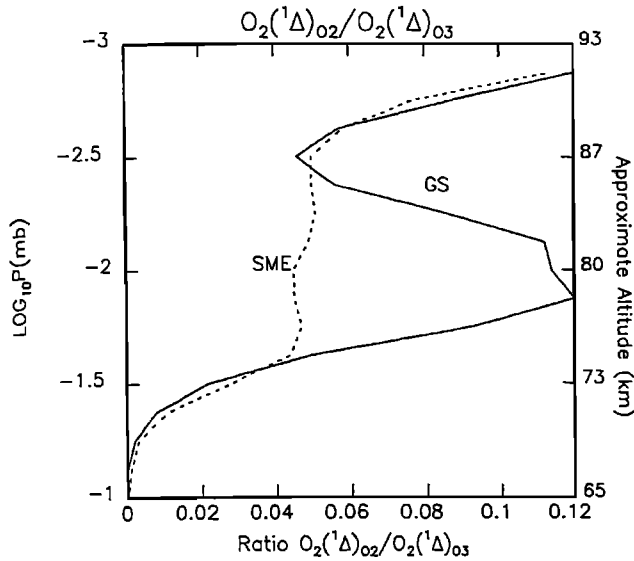


Fig. 9. Comparison of R and R_{SME} in the upper mesosphere and lower thermosphere.

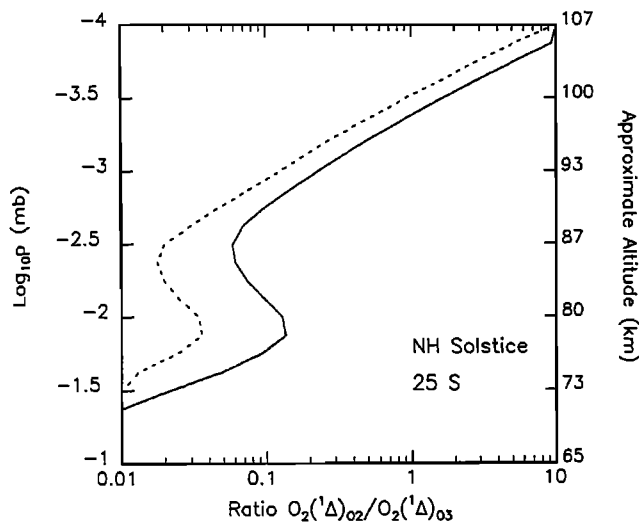


Fig. 10. The parameter R calculated with unit $O(^1D)$ quantum yield from $Ly \alpha$ photolysis and unit energy transfer efficiency from $O(^1D)$ to O_2 (solid curve) compared with R calculated for nonunit yield and efficiency (dashed curve). Values calculated using ozone concentrations from Garcia and Solomon [1983, 1985] two-dimensional model.

the 0.54 transfer efficiency (dashed curve), at solstice. In Figure 10 the $O_2(a^1\Delta_g)$ produced by O_2 photolysis is reduced to a maximum of 1 to 3% of that produced by ozone in the upper mesosphere. In the lower thermosphere, $O_2(a^1\Delta_g)$ produced by O_2 photolysis is still significant, exceeding 5% at 90 km and becoming the dominant source of $O_2(a^1\Delta_g)$ above 100 km. The value of R_{SME} (Figure 11) is found to be less than 1.5% in the upper mesosphere and no more than 5% at 90 km when using nonunit yields and efficiencies, as described above. Consequently, $O_2(a^1\Delta_g)$ resulting from O_2 photolysis may be important only in the lower thermosphere. Figures 10 and 11 clearly illustrate that the quantum yield of $O(^1D)$ over

the $Ly \alpha$ line and the $O(^1D)$ - O_2 transfer efficiency must be more accurately determined at mesospheric temperatures in order that future measurements of $O_2(a^1\Delta_g)$ may properly be interpreted in terms of ozone.

4. Implications for Existing Data Sets

The kinetic model presented in Table 1 and used throughout this paper contains many changes and additions to the kinetic model used in the analysis of SME data. A key question is whether or not the new model implies that significant changes are necessary in the SME O_3 database. It is not directly clear from a comparison of the kinetic models what the effect on the SME database will be, as some changes will tend to increase the retrieved ozone concentrations while others will tend to decrease the retrieved ozone concentrations. For example, the new kinetic model has a smaller Hartley band photolysis rate which will generally imply a larger ozone concentration. However, the new model also has a smaller $O_2(a^1\Delta_g)$ quenching rate (for O_2) which will generally result in smaller ozone concentrations.

To address the question, the following study has been performed using SME ozone data: An SME ozone profile is run through the SME forward kinetic model listed in Table 2 to generate an $O_2(a^1\Delta_g)$ concentration profile. In principle this profile represents the SME-observed $O_2(a^1\Delta_g)$ profile. Using the kinetic model in Table 1 (with ϕ_b and ϕ_c both equal to 1.0), we then calculate the $O_2(a^1\Delta_g)$ concentration for sources other than ozone and subtract them from the SME-observed $O_2(a^1\Delta_g)$ profile, leaving an $O_2(a^1\Delta_g)$ profile that is solely due to ozone photolysis. A new ozone concentration using the ozone-only $O_2(a^1\Delta_g)$ profile and the kinetic parameters in Table 1 is then calculated. Atomic oxygen concentrations from the Garcia and Solomon two-dimensional model are used in calculating quenching of $O_2(a^1\Delta_g)$ and of $O_2(b^1\Sigma_g)$ by O in the new model. The atomic oxygen volume mixing ratio ranges from approximately 10^{-5} at 1.34×10^{-2} mbar (~ 77 km) to 8.0×10^{-3} at 1.34×10^{-3} mbar (~ 92 km).

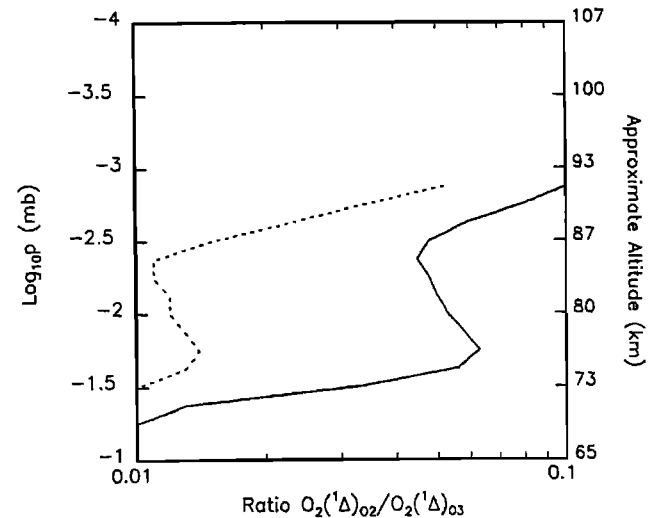


Fig. 11. The parameter R calculated with unit $O(^1D)$ quantum yield from $Ly \alpha$ photolysis and unit energy transfer efficiency from $O(^1D)$ to O_2 (solid curve) compared with R calculated for nonunit yield and efficiency (dashed curve). Values calculated using ozone concentrations from the SME near-infrared spectrometer.

The results of this process are given in Figure 12 which shows the percentage change in the SME ozone concentration as a result of the new kinetic model. Two graphs are shown, one for low latitude (5° N) at solstice and one for midlatitude (40° N) at equinox. As is evident from the figure, application of the new kinetic model will result in a decrease in the amount of ozone in the SME data below 75 km with an increase in the ozone concentration above. The percentage changes are virtually the same at each altitude (pressure level) because, as discussed below, they are dependent on changes in the kinetic parameters and not on the ozone concentration. At 90 km the differences between the ozone changes at 5° N and 40° N are due to the larger atomic oxygen concentration at 40° N which, as discussed below, necessitates a larger ozone concentration.

It is relatively straightforward to explain the shape of the curve in Figure 12. The changes in the inferred ozone profile are driven primarily by changes in the Hartley band photolysis rate and in the quenching rate of $O_2(a^1\Delta_g)$. The ozone concentration retrieved from measurements of the $O_2(a^1\Delta_g)$ volume emission rate can be readily shown to be given by

$$O_3 = \frac{O_2(^1\Delta)_{TOTAL} - O_2(^1\Delta)_{OTHER}}{F} \quad (4)$$

where O_3 is the ozone concentration, $O_2(^1\Delta)_{TOTAL}$ is the $O_2(a^1\Delta_g)$ concentration due to all sources (in this case calculated with the SME forward model), $O_2(^1\Delta)_{OTHER}$ is the $O_2(a^1\Delta_g)$ concentration due to all sources other than ozone, and F is a factor which depends primarily on the ratio of the Hartley band photolysis rate to the $O_2(a^1\Delta_g)$ quenching rate. The ratio of the ozone concentration obtained with the new kinetic model to the SME ozone concentration, O_3^{NEW}/O_3^{SME} , is then given by

$$\frac{O_3^{NEW}}{O_3^{SME}} = \frac{F^{SME}}{F^{NEW}} \frac{O_2(^1\Delta)_{TOTAL} - O_2(^1\Delta)_{OTHER}^{NEW}}{O_2(^1\Delta)_{TOTAL} - O_2(^1\Delta)_{OTHER}^{SME}} \quad (5)$$

where F^{SME} and F^{NEW} are the values of the factor F calculated using the SME parameters and the new model parameters, respectively. The expression on the right-hand side of (5) involving the ratio of $O_2(^1\Delta)$ terms is always slightly less than 1 below 90 km. The two additional sources of $O_2(a^1\Delta_g)$ and the $O_2(a^1\Delta_g)$ resulting from solar excitation of $O_2(b^1\Sigma_g)$ in the new model are, together, greater than or equal to the $O_2(a^1\Delta_g)$ resulting only from solar excitation of $O_2(b^1\Sigma_g)$ alone in the SME model. Consequently, the numerator of the $O_2(^1\Delta)$ expression in (5) is slightly smaller than the denominator resulting in the entire $O_2(^1\Delta)$ term being slightly less than 1. Thus the main cause of differences between O_3^{NEW} and O_3^{SME} lies in the factor F , which can be shown to be proportional to

$$F \propto \frac{J}{A_\Delta + Q_\Delta} \quad (6)$$

where J is the Hartley band photolysis rate, A_Δ is the Einstein coefficient for emission by $O_2(a^1\Delta_g)$ at $1.27 \mu m$, and Q_Δ is the sum of the collisional quenching rates (s^{-1}) of $O_2(a^1\Delta_g)$ for all collision partners. Consequently, the ratio O_3^{NEW}/O_3^{SME} is very nearly given by

$$\frac{O_3^{NEW}}{O_3^{SME}} \equiv \frac{F^{SME}}{F^{NEW}} = \frac{J^{SME}}{J^{NEW}} \frac{A_\Delta + Q_\Delta^{NEW}}{A_\Delta + Q_\Delta^{SME}} \quad (7)$$

The magnitudes of the individual ratios on the right-hand side of (7) are shown in Figure 13. The ratio J^{SME}/J^{NEW} is equal to 9.0/8.1 or 1.1 and is constant with altitude, as shown in the dashed curve labeled "photolysis." A smaller photolysis rate means that more ozone is required to give the observed $O_2(a^1\Delta_g)$ volume emission rate. If changes were made solely to the photolysis rates, the new ozone concentrations would be a factor of 1.1 times larger at all altitudes. However, the $O_2(a^1\Delta_g)$ quenching rates have also changed. The dashed curve labeled "quenching" is the change in the retrieved ozone concentrations solely due to changes in the rate of quenching of $O_2(a^1\Delta_g)$. At low altitudes, $Q_\Delta^{SME} > Q_\Delta^{NEW}$ because the $O_2(a^1\Delta_g)$ quenching is dominated by collisions with O_2 , causing the ratio of the $A_\Delta + Q_\Delta$ terms in (7) to be less than 1. The new and smaller quenching rate increases the lifetime of the $O_2(a^1\Delta_g)$ state, thereby requiring less ozone to produce a specified $O_2(a^1\Delta_g)$ emission rate. If quenching by O_2 was the only $O_2(a^1\Delta_g)$ quenching process considered, the "quenching" curve would asymptotically approach 0% above 80 km. However, we have added quenching of $O_2(a^1\Delta_g)$ by atomic oxygen, which becomes significant above 85 km, resulting in a decrease in the lifetime of $O_2(a^1\Delta_g)$. At high altitudes, $Q_\Delta^{NEW} > Q_\Delta^{SME}$ due to quenching of $O_2(a^1\Delta_g)$ by atomic oxygen (which was not included in the SME model), thereby causing the ratio of the $A_\Delta + Q_\Delta$ term to be greater than 1. To balance the decrease in the lifetime (and hence in the $O_2(a^1\Delta_g)$ concentration), an increase in the ozone concentration is required, as shown above 85 km in the dashed "quenching" curve. The overall effect of changes in the photolysis and quenching rates is given by the sum of the "quenching" and "photolysis" curves and is shown by the solid curve labeled "net" in Figure 13, which is nearly identical to the curves

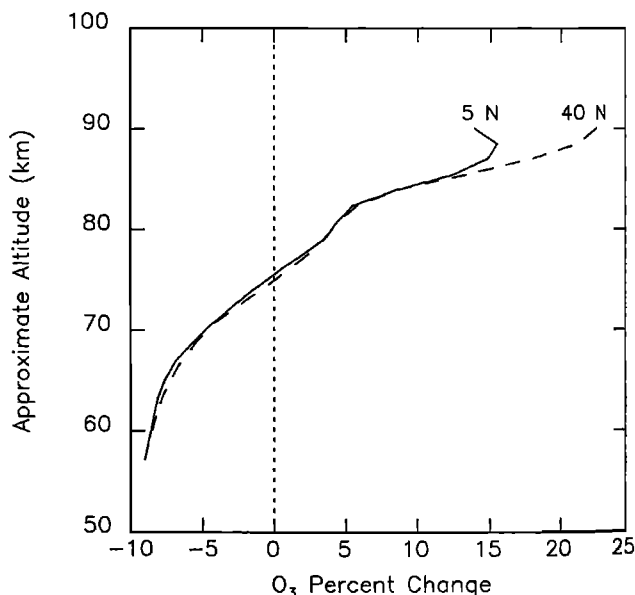


Fig. 12. Percent changes in ozone concentrations derived from SME near-infrared measurements of $O_2(a^1\Delta_g)$ at 5° N (winter) and 40° N (equinox) as a result of changes and updates to the $O_2(a^1\Delta_g)$ kinetic model.

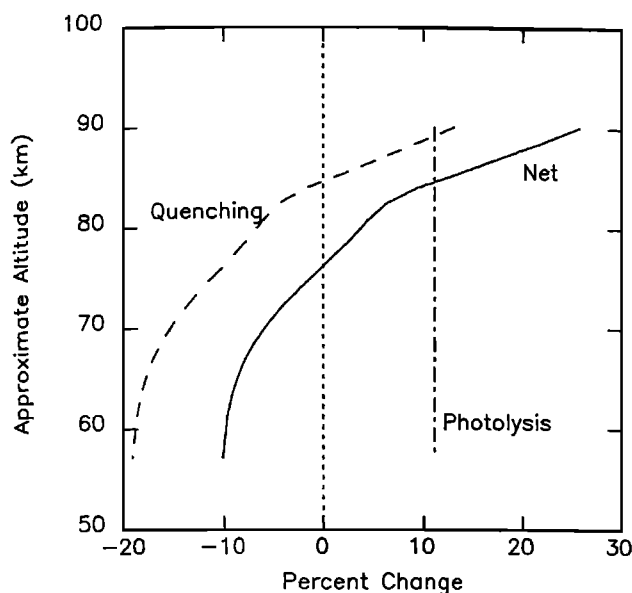


Fig. 13. Percent changes in ozone concentrations derived from SME near-ir measurements of $O_2(a^1\Delta_g)$ due to changes in the $O_2(a^1\Delta_g)$ quenching rate (long-dashed curve), the Hartley band photolysis rate (long-dashed, short-dashed curve). The solid curve is the net change due to the individual changes.

shown in Figure 12. It should be noted that the changes in the ozone concentrations as a function of altitude implied by Figures 12 and 13 are due to changes in the kinetic parameters describing the production and loss of $O_2(a^1\Delta_g)$ and are therefore virtually insensitive to the ozone concentration. The new sources of $O_2(a^1\Delta_g)$ only play a minor role in updating the SME near-infrared ozone concentration retrievals because of the numerous and sometimes compensating changes in the kinetic model.

An important consideration in analyzing measurements of $O_2(a^1\Delta_g)$ emission will be just how sensitive the inferred ozone concentration is to the atomic oxygen concentration used in the retrieval algorithm. To estimate the sensitivity, we have simulated the change in the inferred ozone concentration when the nominal atomic oxygen concentration is changed by $\pm 50\%$, using the new kinetic model. The results are shown in Figure 14. As would be expected, there is very little change in the inferred ozone amount below 80 km due to changes in atomic oxygen because of the relatively small concentration at those altitudes. However, above 80 km a 50% change in the atomic oxygen concentration can change the inferred ozone concentration by 5 to 9%. Clearly, mesopause region ozone abundances inferred from measurements of $O_2(a^1\Delta_g)$ emission will be fairly sensitive to the atomic oxygen concentration used in the retrieval algorithm.

The new $O_2(a^1\Delta_g)$ model described above implies that changes of the order of 10 to 20% are possible in the existing SME near-infrared database. However, the uncertainties in the existing database as a result of kinetics uncertainties most likely limit the accuracy to the 10 to 20% level, so it can be asked if the changes suggested here are physically meaningful. The answer is yes, because the changes in the ozone concentrations implied by the new kinetic model as a

result of changes in the magnitude of the kinetic rates are systematic changes in the inferred concentrations. The magnitude of the kinetic rates governs the magnitude of the retrieved ozone density, while the uncertainty in the kinetic rates governs the size of the retrieval uncertainty (error bars). The size of the error bars will remain virtually the same, but they will span a different range, thus giving a significant change to the range of possible ozone values in the SME database. It should be noted that discrepancies between measured and modeled ozone concentrations are often of the order of 10 to 40%, so systematic changes of 10 to 20% are indeed meaningful.

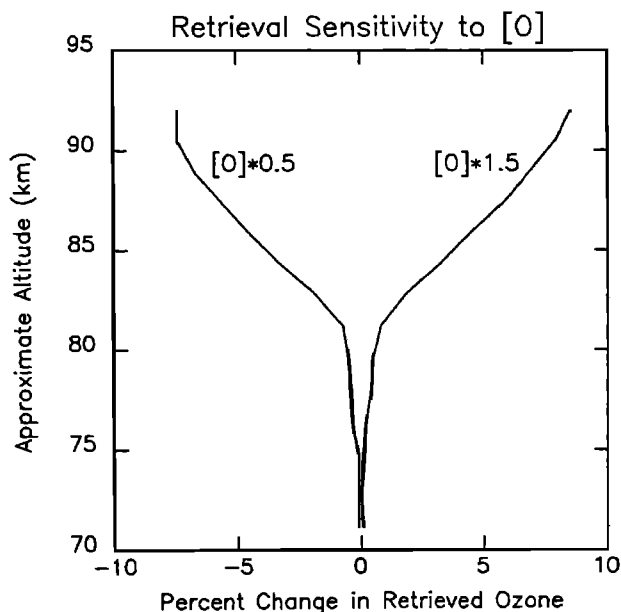


Fig. 14. Sensitivity of the retrieved ozone concentration to a 50% uncertainty in the atomic oxygen concentration. The changes are largest above 80 km where quenching of $O_2(b^1\Sigma_g^-)$ and $O_2(a^1\Delta_g)$ by O becomes significant.

5. Summary and Conclusions

We have demonstrated that $O_2(a^1\Delta_g)$ produced by photolysis of O_2 is a potentially significant source of $O_2(a^1\Delta_g)$ in the upper mesosphere and lower thermosphere. Remote sensing experiments which measure the emission from $O_2(a^1\Delta_g)$ at $1.27 \mu\text{m}$ in order to infer the ozone concentration need to be able to accurately assess the contribution from all sources to the total $O_2(a^1\Delta_g)$ concentration which they observe prior to inferring the ozone concentration. Failure to properly remove these effects will result in spurious temporal (e.g., solar cycle) and latitudinal variations in the database. The calculations which we have presented suggest that Ly α photolysis of O_2 contributes 3 to 15% of the total $O_2(a^1\Delta_g)$ concentration at 80 km and that Ly α and SRC photolysis of O_2 contribute 10 to 15% of the total $O_2(a^1\Delta_g)$ at 90 km at solar maximum. O_2 photolysis in the SRC becomes the dominant source of $O_2(a^1\Delta_g)$ (exceeding that from O_3 photolysis) in the lower thermosphere.

The exact magnitude of the O_2 photolysis contribution to the total $O_2(a^1\Delta_g)$ population depends critically on three factors

which are still in some doubt. First, that the product state of $O_2(b^1\Sigma_g)$ quenching by N_2 , O , CO_2 , and O_2 really is $O_2(a^1\Delta_g)$ and that there is not some partial or complete quenching to the ground state. Second, the quantum yield of $O(^1D)$ from O_2 photolysis at Ly α over the width of the Ly α line must be determined to see if the value of 0.44 reported at line center is representative of the integrated value across the line. This measurement is of utmost importance in quantifying the $O_2(a^1\Delta_g)$ concentration in the upper mesosphere near 80 km. Finally, the uncertainty in the efficiency at which $O(^1D)$ energy is transferred to O_2 (forming $O_2(b^1\Sigma_g)$) must be reduced from the 30% level to perhaps 5 or 10%. All of these quenching parameters and efficiencies must be known at the low kinetic temperatures encountered in the mesosphere and lower thermosphere. Finally, the accuracy to which all relevant kinetic parameters need to be known in order to yield ozone concentrations from future missions accurate to 10 to 15% needs to be determined.

The solar flux variability over the 11-year cycle is believed to be approximately a factor of 2 at Ly α , 50% in the SRC, and a few percent in the Hartley band [Lean, 1991]. Thus the relative contributions of O_2 photolysis to $O_2(a^1\Delta_g)$ concentrations will be greatest for solar maximum (the case considered here) and could be half as large at solar minimum. In addition, short-term variation in Ly α and SRC fluxes over a solar rotation period (27 days) are expected to induce fluctuations in $O_2(a^1\Delta_g)$ concentrations and these sources should be considered in detailed studies of solar activity effects on atmospheric composition. Finally, it is clear that attempts to deduce long-term trends in mesospheric and lower thermospheric O_3 based on $O_2(a^1\Delta_g)$ measurements (e.g., from SME) will be affected by solar-cycle-related changes in the fluxes in these spectral regions.

The updated kinetic model proposed in Table 1 may result in significant changes in the ozone database obtained from the SME NIRS instrument, regardless of the solar cycle. The changes involve a decrease of the order of 10% in the lower mesosphere and an increase of as much as 15 to 20% in the lower thermosphere. The changes are due primarily to changes in the rate of photolysis in the Hartley band of ozone and in the quenching rate of $O_2(a^1\Delta_g)$. In particular, including quenching of $O_2(a^1\Delta_g)$ by atomic oxygen results in an increase in the ozone concentration above 85 km in addition to the increase resulting from the decrease in the Hartley band photolysis rate. The role of atomic oxygen quenching of $O_2(a^1\Delta_g)$ needs to be examined more closely in terms of uncertainties associated with the quenching rate coefficient and with the atomic concentration itself and how these uncertainties impact the accuracy of ozone retrieved from measurements of $O_2(a^1\Delta_g)$ emission in the lower thermosphere.

References

- Braithwaite, M., J. A. Davidson, and E. A. Ogryzlo, $O_2(^1\Sigma)$ relaxation in collisions, I, The influence of long range forces in the quenching by diatomic molecules, *J. Chem. Phys.*, **65**, 771-778, 1976.
- Clark, I. D., and R. P. Wayne, The reaction of $O_2(^1\Delta_g)$ with atomic nitrogen and atomic oxygen, *Chem. Phys. Lett.*, **3**, 405-407, 1969.
- Garcia, R. R., and S. Solomon, A numerical model of the zonally averaged dynamical and chemical structure of the middle atmosphere, *J. Geophys. Res.*, **88**, 1379-1400, 1983.
- Garcia, R. R., and S. Solomon, The effect of breaking gravity waves on the dynamics and chemical composition of the mesosphere and lower thermosphere, *J. Geophys. Res.*, **90**, 3850-3868, 1985.
- Jet Propulsion Laboratory, (JPL) Chemical kinetics and photochemical data for use in stratospheric modeling, in *Evaluation 10*, JPL 92-20, 196 pp., 1992.
- Knickelbein, M. B., K. L. Marsh, O. E. Ulrich, and G. E. Busch, Energy transfer kinetics of singlet molecular oxygen: The deactivation channel for $O_2(b^1\Sigma_g)$, *J. Chem. Phys.*, **87**, 2392-2393, 1987.
- Lean, J., Variations in the sun's radiative output, *Rev. Geophys.*, **29**, 505-535, 1991.
- Lee, L. C., and T. G. Slanger, Observations of $O(^1D \rightarrow ^3p)$ and $O_2(b^1\Sigma_g \rightarrow X^3\Sigma_g)$ following O_2 photodissociation, *J. Chem. Phys.*, **69**, 4053-4060, 1978.
- Lee, L. C., T. G. Slanger, G. Black, and R. L. Sharpless, Quantum yields for the production of $O(^1D)$ from photodissociation of O_2 at 1160-1770 Angstroms, *J. Chem. Phys.*, **67**, 5602-5606, 1977.
- National Aeronautics and Space Administration, (NASA), Thermosphere-ionosphere-mesosphere energetics and dynamics (TIMED), report of the Science Definition Team, vol. II, Off. of Space Sci., Washington, D. C., 1991.
- Ogryzlo, E. A., and B. A. Thrush, The vibrational excitation of H_2O and CO_2 by $O_2(^1\Sigma_g^+)$, *Chem. Phys. Lett.*, **24**, 314-316, 1974.
- Rothman, L. S., et al., The HITRAN database: 1986 edition, *Appl. Opt.*, **26**, 4058-4097, 1987.
- Sica, R. J., Inferring middle-atmospheric ozone height profiles from ground-based measurements of molecular oxygen emission rates, I, Model description and sensitivity to inputs, *Can. J. Phys.*, **69**, 1069-1075, 1991.
- Thomas, R. J., C. A. Barth, D. W. Rusch, and R. W. Sanders, Solar Mesosphere Explorer near-infrared spectrometer: Measurements of 1.27- μm radiances and the inference of mesospheric ozone, *J. Geophys. Res.*, **89**, 9569-9580, 1984.
- Wildt, J., G. Bednarek, and E. H. Fink, Laser excitation of $O_2(b^1\Sigma_g, v' = 0, 1, 2)$ rates and channels of energy transfer and quenching, *Chem. Phys.*, **122**, 463-470, 1988.
- World Meteorological Organization (WMO), The stratosphere 1981 theory and measurements, *Report No. 11*, Global ozone research and monitoring project, 1981.
- World Meteorological Organization (WMO), Report of the international ozone trends panel, *Report No. 18*, Global ozone research and monitoring project, vol. 1, 1988.
- Martin G. Mlynczak, NASA Langley Research Center, Mail Stop 401B, Hampton, VA 23681-0001.
- Susan Solomon, NOAA Aeronomy Laboratory, 325 S. Broadway, Boulder, CO 80303.
- Daphne S. Zaras, Science Applications International Corporation, Hampton, VA 23666.

(Received February 20, 1993;
revised June 1, 1993;
accepted June 1, 1993.)